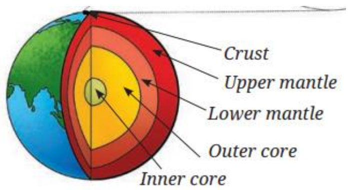
Chapter – 13: Our Home: Earth, a Unique Life Sustaining Planet

- Reached – final chapter – this book – concluding chapter – scientific journey – middle stage
- Time – put together – all – learnt – try – understand – why – earth – no duplicate – entire universe?
- Learnt – curiosity – grade 6, 7 – earth – planet – orbit sun
- BUT – not any planet – this planet – sustains life – diverse landscapes – towering mountains – vast oceans – endless deserts – lush forests
- Today – satellites – allow – amazing pictures – planet
- Img. – here – taken – International Space Research Organisation (ISRO) – Earth Obs. Satellite – made – combine – 3000 smaller img.
- This img. – very beautiful – BUT – false colour img. – scientists use – diff. colour – diff. info.
- Satellite img. – help study – plants – land – tiny org. – oceans – detect – ocean temp., oil spill, wind dir.
- This chapter – uncover – unique cond. – make earth – perfect home – living beings

Why is Earth a Unique Planet?

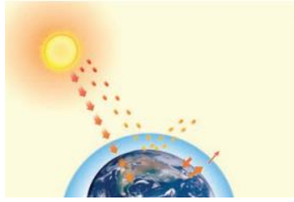
- Billions – planets – universe – earth – only one – life exists
- Ever wonder – all life – earth – actually exist?
- Mountains, rivers, forests, animals, people – found – very thin layer – surface – planet
- Tallest mountain – deepest trench – crust – all life exist – tiny – compare – entire earth



- Earth – apple size – crust – apple's skin
- Activity –
  - List – features – earth – often take – granted – BUT – interesting, imp.
  - Note them – table
- Discuss features – listed – teacher, friends
- May realise – earth – interesting, imp. – many ways
- Provides – air – breathe – water – drink – soil – help – grow crops
- Earth – also provide – materials – rock, timber – make homes, buildings, roads
- You – curious – know – earth – unique planet – how?

What Do The Planets of Our Solar System Look Like?

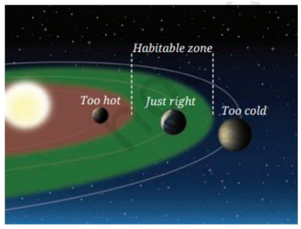
- Curiosity – grade 6 – studied – solar sys. – chapter – beyond earth
- Recall – some things – learnt
- Solar sys. – 8 planets – revolve – sun – nearly circular orbits
- Order – increasing dist. – Sun – Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune
- 1st 4 – rel. small – rocky – rest – large – made – gases
- Activity –
  - **Collect** – info. – temp., size – planets – solar sys. – check – atm.
  - Collect info. – books – school library, websites, discuss – teachers
  - Fill data – table
- All planets – solar sys. – energy – sun
- Planet – very close – sun – very hot – move away – planets – colder



- This – gen. correct – except – venus – 2nd planet – sun – highest avg. temp. – hottest planet
- Venus – hottest – reason – thick atm. – trap heat
- Air – venus – almost CO_2 – no escape – heat – **greenhouse effect**
- Earth – gases – CO_2 – trap heat – absorb radiation – after getting warm
- Greenhouse effect – imp. role – maintain temp. – earth
- **A Step Further**
  - Greenhouse effect – allow planets – venus, earth – trap heat – not same – greenhouse – grow crops – cool climate
  - Venus, earth – gases – CO_2 – atm. – trap heat – absorb radiation – given off – gets warm – sun
  - Plant greenhouse – traps – warm air – closed space – glass walls – heat – daytime – air – stays in – heat no escape
  - Both – keep warm – differently

What Makes the Earth Surface Suitable for Life to Exist?

Position of the earth



- Most imp. reason – earth – support life – dist. – sun
- Dist. – just right – temp. – allow water – exist – liquid form
- Earth – closer – sun – too hot – water – evaporate
- Earth – farther away – sun – too cold – water – freeze
- Extreme cond. – impossible – most life forms – plants, animals, humans – grow, thrive – earth
- Some microbes – certain bacteria – survive – frozen env.
- Earth dist. – sun – allow water – liquid form – essential – development, sustenance – life – all forms
- Range – dist. – sun (any star) – water – liquid form – **habitable zone** – also – **Goldilocks zone**
- Also studied – Social Science – most surface – earth – covered – water
- Thus – viewed – space – earth – blue – reason – vast amt. – water – hence – blue planet
- **A Step Further**
  - Mars – edge – sun – habitable zone
  - Several spacecraft – sent – mars – rovers – land, explore surface – no life found
  - Scientists believe – past – mars – liquid water – lakes , cond. – support simple life
  - This reason – 1 of many – mars – interest scientists
  - Also remind – science – not always – final answer
  - Explore more – new clues – even new life
  - Science – stay open – change – learn more

Size of the earth

- More imp. factors – make earth – habitable
- Solar sys. – orbits – most planets – almost circular
- Keeps amt. – sunlight, heat – steady – throughout yr. – prevent – extreme summers, winters – most places
- BUT – moderate temp. – not the only factor – earth – habitable
- Planet – also – right size – support atm.
- Learnt earlier – atm. – layer – gases – surround earth – major role – sustaining life
- Also learnt – chapter 5 – earth's gravity – pull obj. – toward itself

- Earth – much smaller – same density – gravity – too weak – hold – gases – atm.
- Mars – atm. – 100x thinner – mercury – not atm.
- Other hand – planet – bigger – gravity – stronger – pull down – people – so hard – bones – crush
- Right size – reason – support atm. – essential life
- Presence – oxygen – atm. – allow – breathing – needed – all forms – life
- Oxygen – another imp. role – some oxygen – convert to – ozone (O_3) – imp. part – atm. – **ozone layer**
- This layer – shield – block – ultraviolet (UV) rays – sun – damage living cells
- **Important fact** –
  - India – Mangalyaan (Mars Orbiter Mission) – 2013 – ISRO – big step – explore mars
  - Carry tools – study – atm., surface, signs – past water
  - Some sensors – help scientists – big ques. – mars was ever habitable?
  - Mangalyaan – proved – India – space science – smart, low-cost tech. – help bring mars close

Magnetic field of the earth

- Curiosity – grade 6 – learnt – freely suspended magnet – settle – fixed dir.
- Reason – earth – behave – giant magnet
- Also seen – chapter 4 – region – around magnet – effect felt – magnetic field
- Believed – movement – molten iron – earth's core – origin – earth's magnetic field
- Earth – constantly hit – tiny, high-energy particles
- Some – far universe – **cosmic rays** – others – sun – **solar wind**
- These particles – harmful – damage env. – reduce – ozone layer – let in – harmful UV rays – affect life – earth
- Earth's magnetic field – work – protective shield – push away – harmful particles – keep atm., planet – safe
- Earth's unique position – solar sys. – allow presence – liquid water – along – size, atm., magnetic field – all help – make possible – life – emerge, thrive

What Allows Life to Be Sustained on Earth?

- Earth – right condition – life – BUT – beautiful connections – living, non-living things – help life – thrive
- Curiosity – grades 6, 7 – learnt – key natural resources – air, water, sunlight, minerals, soil
- Also learnt – imp. life processes – plants, animals
- Now – **explore** – all elements – interact – support, sustain life

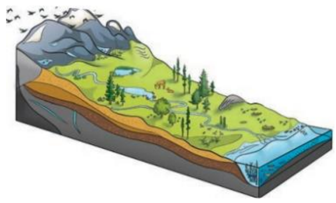
Air, water, and sunlight

- We know – atm. – oxygen – humans, animals, plants – use – respiration
- Presence – sunlight – plants – take in CO_2 from air – water from soil – prepare food – photosynthesis
- This process – O_2 released – needed – respiration
- Learnt – radiation – sun – heats earth
- Some heat – trapped – atm. – greenhouse effect
- This effect – mild – BUT – keeps temp. – high enough – water – remain liquid
- Curiosity – grade 7 – also learnt – heat transfer – radiation
- w/o atm. – earth – lose heat – become cold – greenhouse effect – keeps earth warm
- Water – essential – life

- Learnt – covers – 70 % - earth surface – found – ponds, lakes, reivers, springs, seas, oceans, groundwater
- All this water – **hydrosphere**
- Curiosity – grade 7 – learnt – water – good solvent
- Also learnt – water – transport nutrients – plants
- Animals – water – regulate – body temp. – air digestion – ensure hydration
- Much earth – covered – water – little knowledge – deep oceans
- Hydrosphere – home – millions – life forms – tiny planktons to giant whales
- Oceans, lakes, rivers – rich env. – aquatic life – freshwater – also needed – grow crops – support people
- Water vapour – air – form clouds – make rain, snow – refill rivers, lakes, underground water
- Rainfall – affect all – plants, animals – live – specific place
- Moving air – shape weather, rainfall – influence farming, water supply, life

Soil, rocks, and minerals

- Beneath feet – something remarkable – earth's crust – rocks, soil, minerals
- Looks – hard, lifeless – provides everything – life needs – grow, survive
- Soil – help plants – grow – minerals – salt, coal, oil, metals – iron, copper – outer layer – support both – ecosys., human life
- Solid part – earth – inc. materials – rocks, soils, minerals – **geosphere**
- Soil – looks – simple dirt – BUT – rich – nutrients – nitrogen, potassium – plants need – grow
- These nutrients – result – slow breakdown – rocks – remains – plants, animals
- Various kinds – landforms, soils, rocks – this variety – processes – alter them – **geodiversity**
- Helps create – unique habitats – diff. life – thrive
- Nature – non-living parts – soil, rocks, water – help shape – life



Plants, animals, microorganisms

- Chapter – microbes – chapter – ecology – seen earth – full – life – trees, shrubs, herbs – animals, insects, tiny org. – invisible – naked eye
- Living beings – places – they live – **biosphere** – inc. – land, air, water – life – interact – surroundings – survive, grow
- Learnt – chapter 12 – living beings – depend – each other, env.
- Plants – own food – photosynthesis – animals – eat – plants, animals – decomposers – break down – dead matter – return nutrients – soil
- Nature – work together – sys. – support life

The importance of balance

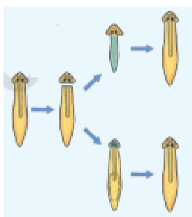
- Earth – giant teamwork project – nature, weather, life
- Vast, living sys. – land, air, water, living things – support, affect – each other
- Small change – 1 part – cutting forest – impact – rainfall, soil, air quality, animals
- Life – earth – survive – everything – work together – balance
- This balance – keeps planet habitable
- Preserving, protecting – clean air, water, soil, all life forms – imp., essential – keep earth healthy – future

What Keeps Life from Disappearing?

- Plants, animals – no reproduction – life – disappear – earth
- Reproduction – ensure – each org. – exist – maintain life
- Usually expect – animals – produce young ones – resemble them – cows – calves – cats – kittens
- Parents – pass on instructions – offsprings – develop – single cell
- These instructions – **genetic materials** OR **genes** – stored – every cell – living being
- Genes – detailed manual – each cell
- Some instructions – tell cell – make blood – others – guide formation – bones, muscles, skin
- Together – instructions – ensure – calves to cows – kittens to cats
- Reproduction – create same kind – BUT also – allow – some changes – instructions
- Sometimes – changes – help plants, animals – survive better – new env.
- Over time – camels – develop humps – store fat – survive – deserts
- Microbes – some bacteria – learnt – chapter – health – become resistant – antibiotics
- Over gen. – such changes – lead – new features – sometimes – completely new kind – org.
- Same process – both – similarity, variations – how?
- 2 types – reproduction
    - One – young ones – almost same – parents
    - Another – slightly diff. – parents
- Asexual reproduction – single parent – new individual – exact copies
- Sexual reproduction – instructions – 2 parents – combine – offspring – not exactly same
- Share traits – both parents – also some diff.
- Mixing – help – keep useful features – let new appear
- Many gen. – diff. – add up – lead – big diff. – even new life forms

Asexual reproduction

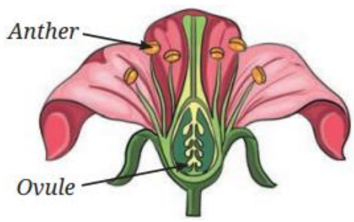
- Many plants – reproduce – any part – leaf, stem, root – planted – soil
- This kind – reproduction – **vegetative propagation**
- Activity –
    - Take – some part – plant – stem – money plant – eyes – sprouted potato – piece – ginger
    - Plant each – separately – moist soil – money plant – use glass container – easy obs.
    - Make sure – all meet cond. – grow – water, air, sunlight
    - Watch – every day – note – time taken – other parts – appear – also obs. – 1st new leaf
- **Interesting fact** –
    - Not only plants – microbes, simple animals – also reproduce – asexually
    - Ex. – single-celled org. – bacteria, amoeba – divide – 2 identical individuals
    - Some multicellular org. – algae – regrow – small parts
    - Hydra – grow – tiny buds – break off – grow – new org.
    - Planaria – flatworm – regrow – fragment – body
    - Scientists – study planaria – understand regeneration



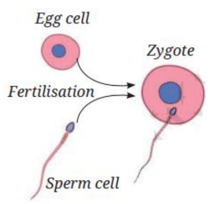
Sexual reproduction

- This type – reproduction – 2 parents – involved – male, female
- Easy – obs. – animals – BUT – flowering plants – male, female parts
- Some microbes – bacteria, yeast – 2 mating types – act – 2 parents

- **Special cells for reproduction**
  - Might wonder – both parents – pass info. – new org. – child – double amt. – info.? – doubles – every gen.?
  - Not happening – reason – each parent – special cells – **gametes** – carry half genetic materials – parents
  - Male, female gametes – join – new cell – complete set – genetic info. – half – each parent
  - Babies – no exact copy – mother, father – brothers, sisters – look diff. – same family
  - Reason – baby – mix – genetic info. – both parents
  - Each gamete – diff. set – info. – things – eye colour, hair type, etc.
  - This info. – mix – diff. ways – sperm, eggs – join – make baby – each child – unique
  - All depend – which piece – parents' info. – came together
- **Sexual reproduction in plants**
  - Plants – diff. parts – flower – produce – male, female gametes
  - Pollen grains – anther – male gametes – ovules – female gametes
  - Pollen – carried – another flower – wind, insects, animals – process – **pollination**
  - Male, female gametes – combine – **fertilisation** – form zygote – become seed



  - Fleshy part – flower – around ovule – develop – fruit
  - Birds, animals – eat fruit – seeds – dropped far – original plant – helpful way – plants – spread
  - Banyan seed – dropped – bird – might sprout – wall crack – after rains
  - Seed – gets water – use stored nutrients – grow – roots, shoots
  - Remember – grade 6 – studied germination – seeds – obs. – tiny shoots, 1st leave
- **Sexual reproduction in animals**
  - Animals – gametes – sperm (male) – eggs (female)
  - Fertilisation – take place – water – ex. – male, female – fish, frogs – eject sperm, eggs – water – combine – form zygote
  - Birds, mammals – inc. humans – sperm deposit – inside body – female – fertilisation – sperm swim – female egg



  - After this – birds, mammals – diff. processes
  - Birds – fertilized eggs – laid – female – development – embryo – hatching process
  - This – 1 strategy – ensure – nutrition supply – embryo
  - Most mammals – development – embryo – inside female body
  - Mother's body – provide nutrition – baby needs – grow
  - This diff. way – giving nutrition – developing baby –

What Are the Threats to Life on Earth?

- We know – life – earth – depends – delicate balance – living, non-living things – work together
- BUT – human actions – disturb balance – small changes – global temp. – oxygen levels – ozone layer – put life – risk
- Today – biggest challenge – climate change, biodiversity loss, pollution – together – triple planetary crisis
- Burning fossil fuels – coal, oil – release – greenhouse gases – CO_2, methane
- Trap – more heat – cause – global warming
- Earth – keep balance – trees, plants, tiny ocean planktons – absorb CO_2

- BUT – burn fossil fuel – release extra carbon – locked – millions – yrs.
- Earth – can't absorb – fast enough – heat – build up
- Small increase – temp. – melt ice caps – raise sea levels – flood – coastal cities – extreme weather cond.
- Long-term changes – temp., rainfall, weather patterns – **climate change**
- Natural habitats – destroyed – plants, animals – disappear – upset ecosys.
- Ex. – chapter 12 – grass vanish – animals – deer, grasshoppers – struggle – survive – predators – tigers, foxes – lose food too
- Every type – living thing – some role – losing few – weaken – nature's ability – support life
- Pollution – add – problem
- Air pollution – factories, vehicles, burning fuels – harm both – nature, people
- Cause – breathing problems – damage crops – lead – smog, acid rain
- Climate change – affect everything – crop growth, water supply, wildlife, human health
- Protect life – earth – cut pollution – use clean energy – make wiser choices
- Also learnt – life – earth – flourish – delicate balance – interdependent natural sys.
- BUT – this balance – threatened – human actions
- Ex. – less or more – oxygen, temp., ozone – endanger life – earth
- **Interesting fact** –
    - Countries – global agreements – protect env.
    - Montreal Protocol (1987) – reduce chem. – Chlorofluorocarbons (CFCs) – allow ozone layer – recover
    - Earth Summit (1992) – international efforts – climate change, biodiversity
    - Kyoto Protocol (2005), Paris Agreement (2015) – commit countries – reduce greenhouse gas emissions
    - Paris Agreement – goal – limit global warming – $< 1.5\ ^{\circ}C$ – latest 2025 – word – not on track
    - Much more action – needed – avoid further damage
    - Water, soil pollution – serious threats – life
    - Factory, farm, plastic waste – harm – aquatic life
    - Excessive fertilisers – poor waste disposal – pollute soil – reduce crop yield – spread harmful substance – food chain
    - Protect them – req. – better waste mgmt. – sustainable farming
    - Seen above – earth sys. – hydrosphere, biosphere, atm., geosphere – connected – damage – 1 – affect others
    - Protect climate – cut down – greenhouse gases – renewable energy – solar, wind – improve energy use – choose env. friendly ways – travel
    - Same time – preserve biodiversity – key – diverse ecosys. – stronger, more balanced
    - Local comm. – big role – sustainable use – natural resources
    - Everyone – help – reuse, repair, recycle – clothes, plastics – reduce pollution, waste
    - Small actions – save energy, water – add up
    - Learn more – share ideas – encourage others – make diff.
    - **Conclusion** – sustain life – earth – need action – everyone
    - Work together – live responsible – protect unique planet – future